

Bridging Sim-to-Real Gap in Event-Driven Reinforcement Learning for Autonomous Navigation

Isaac Van Benthuyssen, Bijay Gaudel (PhD student)
Department of Mechanical Engineering, Advisor Hamid Jafarnejad Sani

Introduction

Sim-to-Real Transfer

- Training in simulation is easier, faster, safer, and cheaper
- However, often considerable performance drop when transferring simulation-trained models to real world

Event Camera

- Novel type of sensor that records independent changes in pixel brightness, rather than entire frames

Reinforcement Learning

- Trains an algorithm to make the correct decisions to achieve a goal, via a trial-and-error process

Goal of Study

The goal of this study is to make improvements to the training simulation that will better the real-world performance of the algorithm.

Methods

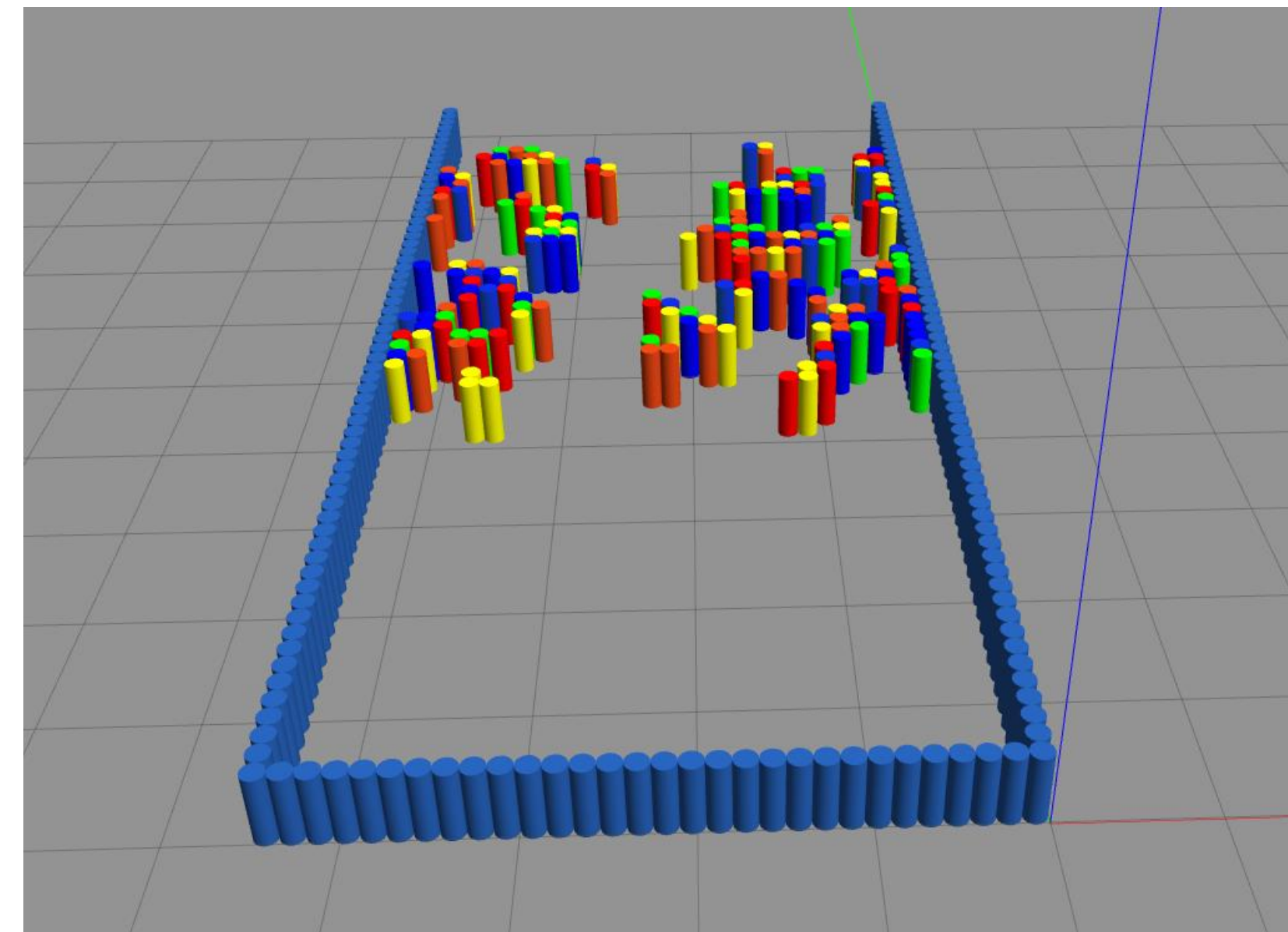
Creating More Realistic Worlds

- Previous worlds consisted of smooth, single-colored shapes
- Worlds with textured shapes and random objects expose the algorithm to a variety of real-world textures and items
- Environments textured with the walls and floors of our real-world testing area and populated with cardboard box obstacles allow the algorithm to train in a highly realistic environment

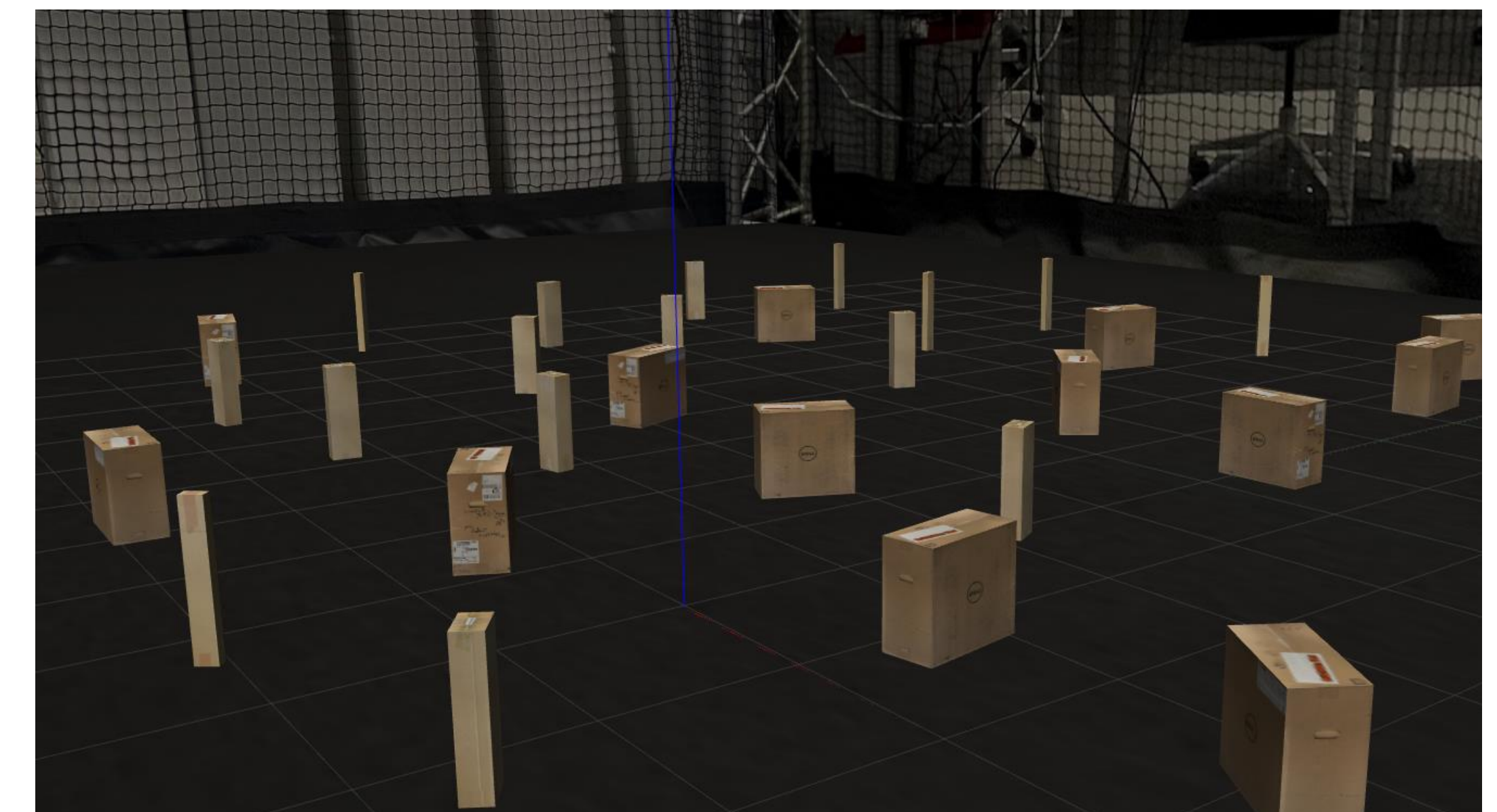
Ensuring Realism of Event Simulator

- The algorithm trains with a simulated event camera, and an inaccurate event simulator would widen the sim-to-real gap
- Lab replica worlds allowed us to compare real and simulated event frames taken from the same environment

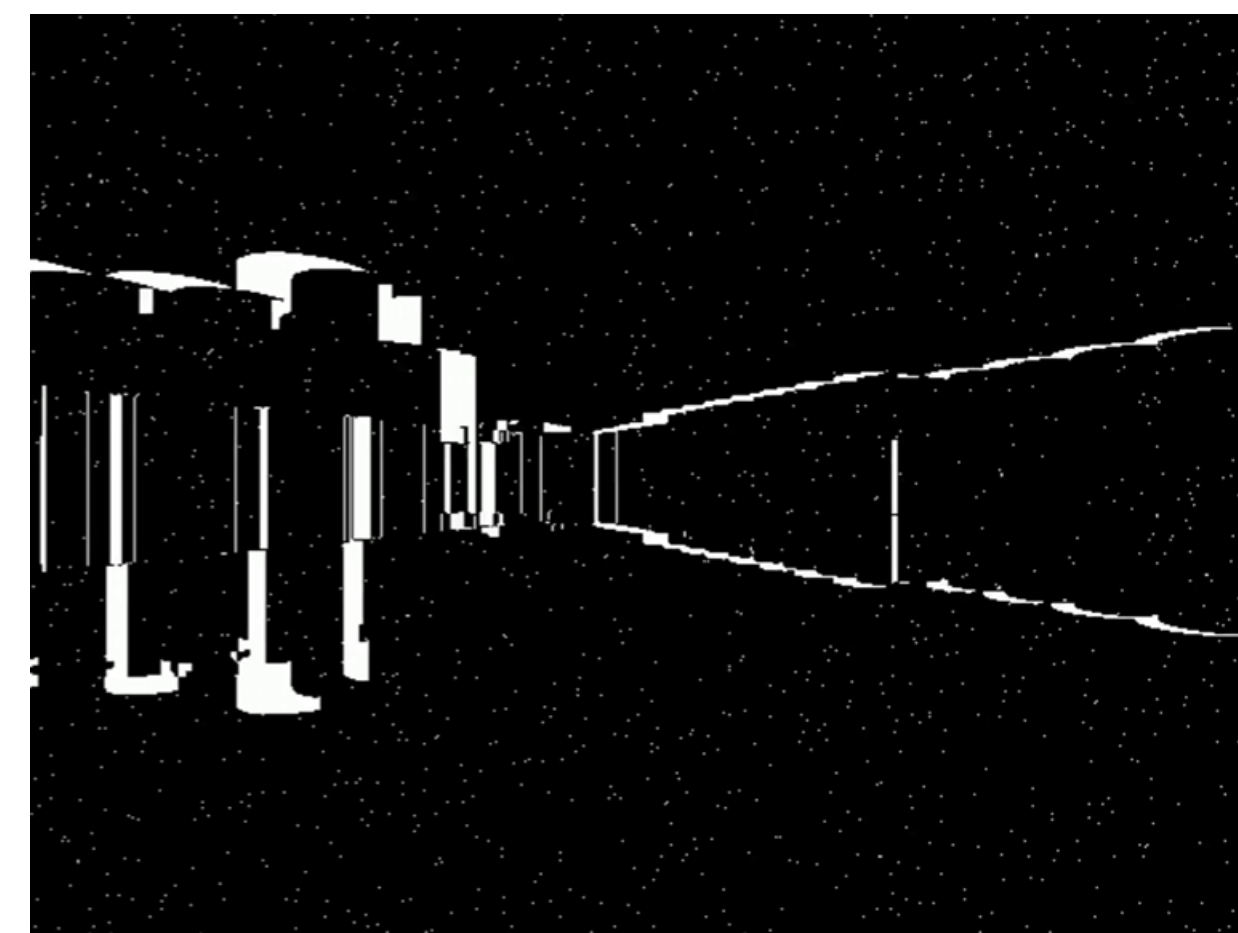
Results



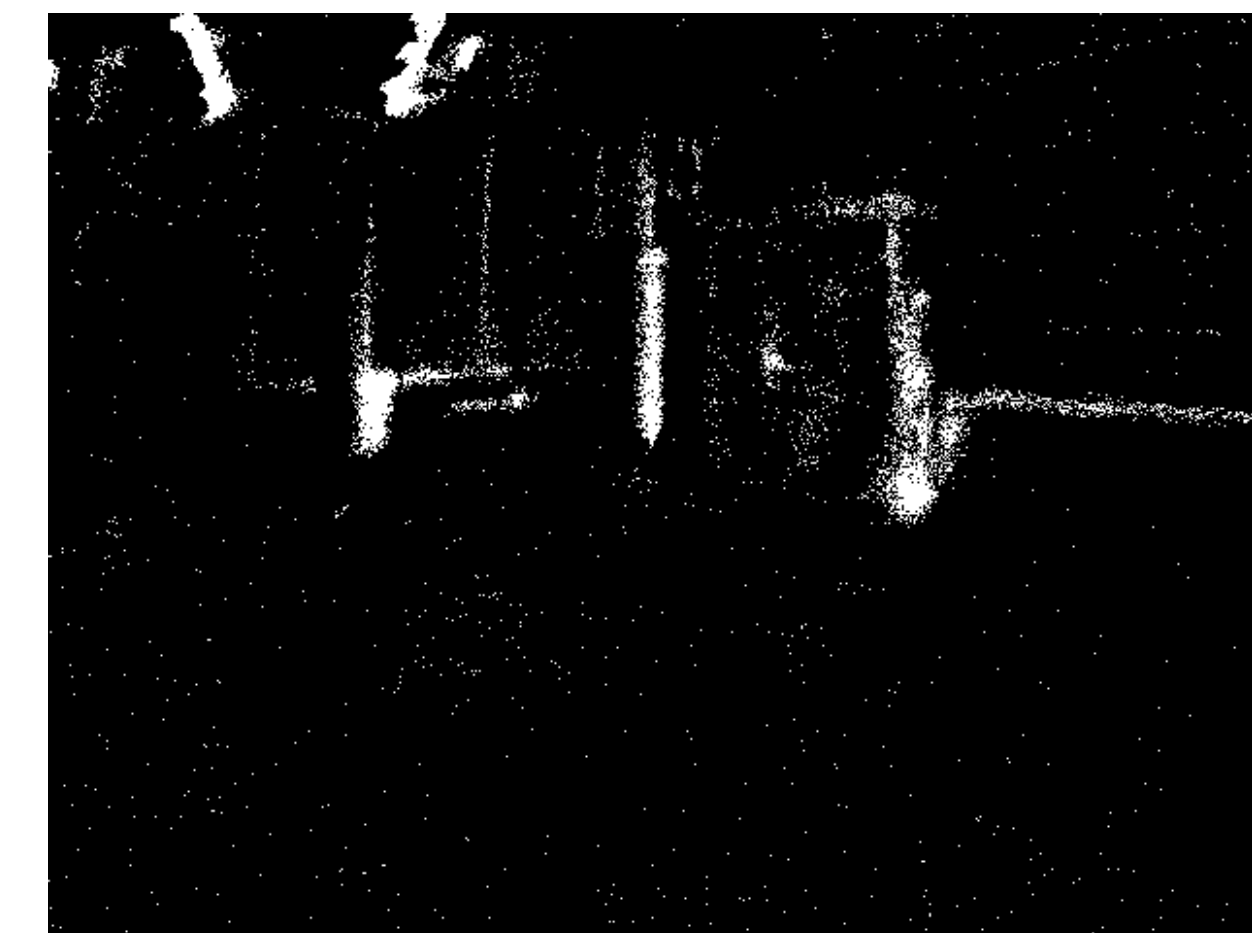
Old world



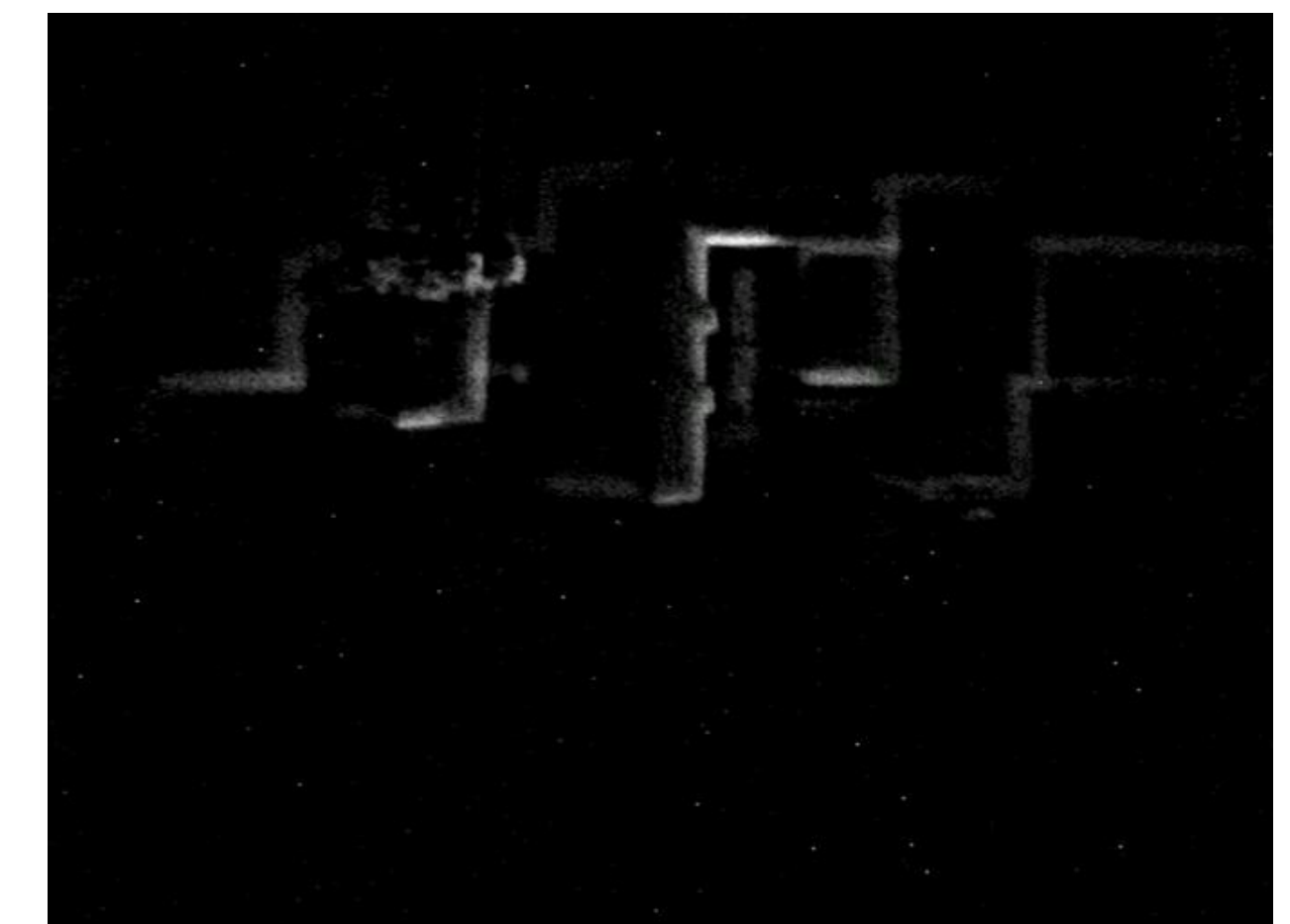
New world



Simulated events in old worlds



Simulated events in new worlds



Real events in real world

- Realistic simulation worlds produce event frames very similar to real-world event frames
- With previous worlds: 6/10 successful navigations in simple obstacle courses
- With new worlds: Tests to be conducted

Conclusions & Future Prospects

- Increasing realism of simulation worlds appears to be simple way to bridge sim-to-real gap
- Future steps:
 - Complete training of reinforcement learning models
 - Conduct real-world tests
 - Compare performance when trained on old vs. new worlds